Mathematical Analysis II
Week 9 Homework (April 28)

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Section 10.3

1. 计算下列三重积分.

(2) $\displaystyle\iiint_V xy^2z^3\,\mathrm{d}x\mathrm{d}y\mathrm{d}z$, V: 由 $z=xy$, $y=x$, $x=1$, $z=0$ 围成;

解. 该区域位于第一卦限, 积分区域可表示为 $0\le x\le 1$, $0\le y\le x$, $0\le z\le xy$.

因此

$$\begin{aligned}\iiint_V xy^2z^3\,\mathrm{d}x\mathrm{d}y\mathrm{d}z &= \int_0^1 \mathrm{d}x\int_0^x \mathrm{d}y\int_0^{xy} xy^2z^3\,\mathrm{d}z\\ &= \frac{1}{4}\int_0^1 \mathrm{d}x\int_0^x x^5y^6\,\mathrm{d}y\\ &= \frac{1}{28}\int_0^1 x^{12}\,\mathrm{d}x\\ &= \frac{1}{364}\end{aligned}$$

(4) $\displaystyle\iiint_V (a-y)\,\mathrm{d}x\mathrm{d}y\mathrm{d}z$, V: 由 $y=0$, $z=0$, $2x+y=a$, $x+y=a$, $y+z=a$ 围成.

解. 积分区域满足 $0\le y\le a$, $\dfrac{a-y}{2}\le x\le a-y$, $0\le z\le a-y$.

因此

$$\begin{aligned}\iiint_V (a-y)\,\mathrm{d}x\mathrm{d}y\mathrm{d}z &= \int_0^a \mathrm{d}y\int_{\frac{a-y}{2}}^{a-y}\mathrm{d}x\int_0^{a-y}(a-y)\,\mathrm{d}z\\ &= \int_0^a \mathrm{d}y\int_{\frac{a-y}{2}}^{a-y}(a-y)^2\,\mathrm{d}x\\ &= \frac{1}{2}\int_0^a (a-y)^3\,\mathrm{d}y\\ &= \frac{a^4}{8}\end{aligned}$$

2. 计算下列积分值.

(2) $\displaystyle\int_{-R}^{R} \mathrm{d}x \int_{-\sqrt{R^2-x^2}}^{\sqrt{R^2-x^2}} \mathrm{d}y \int_{0}^{\sqrt{R^2-x^2-y^2}} (x^2+y^2)\,\mathrm{d}z$;

解. 采用柱面坐标变换: 记 $\begin{cases} x = r\cos\theta \\ y = r\sin\theta \\ z = z \end{cases}$, 则积分区域为 $\begin{cases} 0 \le r \le R \\ 0 \le \theta \le 2\pi \\ 0 \le z \le \sqrt{R^2-r^2} \end{cases}$, 雅可比行列式为 $\dfrac{\partial(x,y,z)}{\partial(r,\theta,z)} = r$.

因此

$$\begin{aligned}
\int_{-R}^{R} \mathrm{d}x \int_{-\sqrt{R^2-x^2}}^{\sqrt{R^2-x^2}} \mathrm{d}y \int_{0}^{\sqrt{R^2-x^2-y^2}} (x^2+y^2)\,\mathrm{d}z &= \int_0^R \mathrm{d}r \int_0^{2\pi} \mathrm{d}\theta \int_0^{\sqrt{R^2-r^2}} r^3\,\mathrm{d}z \\
&= \int_0^R \mathrm{d}r \int_0^{2\pi} r^3\sqrt{R^2-r^2}\,\mathrm{d}\theta \\
&= 2\pi \int_0^R r^3\sqrt{R^2-r^2}\,\mathrm{d}r
\end{aligned}$$

对 $\displaystyle\int r^3\sqrt{R^2-r^2}\,\mathrm{d}r$ 采用分部积分法. 记 $u = r^2$, $\mathrm{d}v = r\sqrt{R^2-r^2}\,\mathrm{d}r$, 则 $\mathrm{d}u = 2r\,\mathrm{d}r$, $v = -\dfrac{1}{3}(R^2-r^2)^{\frac{3}{2}}$.

因此

$$\begin{aligned}
\int r^3\sqrt{R^2-r^2}\,\mathrm{d}r &= \int u\,\mathrm{d}v \\
&= uv - \int v\,\mathrm{d}u \\
&= -\frac{r^2}{3}(R^2-r^2)^{\frac{3}{2}} + \frac{2}{3}\int r(R^2-r^2)^{\frac{3}{2}}\,\mathrm{d}r \\
&= -\frac{r^2}{3}(R^2-r^2)^{\frac{3}{2}} - \frac{2}{15}(R^2-r^2)^{\frac{5}{2}} + C
\end{aligned}$$

从而

$$\begin{aligned}
\int_0^R r^3\sqrt{R^2-r^2}\,\mathrm{d}r &= \left(-\frac{r^2}{3}(R^2-r^2)^{\frac{3}{2}} - \frac{2}{15}(R^2-r^2)^{\frac{5}{2}}\right)\Bigg|_0^R \\
&= \frac{2}{15}R^5
\end{aligned}$$

因此

$$\begin{aligned}
\text{原式} &= 2\pi \int_0^R r^3\sqrt{R^2-r^2}\,\mathrm{d}r \\
&= \frac{4}{15}\pi R^5
\end{aligned}$$

(4) $\int_0^1 \mathrm{d}x \int_0^{\sqrt{1-x^2}} \mathrm{d}y \int_{\sqrt{x^2+y^2}}^{\sqrt{2-x^2-y^2}} z^2 \,\mathrm{d}z$.

解. 该区域位于第一卦限.

采用柱面坐标变换: 记 $\begin{cases} x = r\cos\theta \\ y = r\sin\theta \\ z = z \end{cases}$ ，则积分区域为 $\begin{cases} 0 \le r \le 1 \\ 0 \le \theta \le \frac{\pi}{2} \\ r \le z \le \sqrt{2-r^2} \end{cases}$ ，雅可比行列式为 $\dfrac{\partial(x,y,z)}{\partial(r,\theta,z)} = r$.

因此

$$\begin{aligned}
\int_0^1 \mathrm{d}x \int_0^{\sqrt{1-x^2}} \mathrm{d}y \int_{\sqrt{x^2+y^2}}^{\sqrt{2-x^2-y^2}} z^2 \,\mathrm{d}z &= \int_0^1 \mathrm{d}r \int_0^{\frac{\pi}{2}} \mathrm{d}\theta \int_r^{\sqrt{2-r^2}} rz^2 \,\mathrm{d}z \\
&= \frac{1}{3}\int_0^1 \mathrm{d}r \int_0^{\frac{\pi}{2}} r\left[(2-r^2)^{\frac{3}{2}} - r^3\right] \mathrm{d}\theta \\
&= \frac{\pi}{6}\int_0^1 \left[r(2-r^2)^{\frac{3}{2}} - r^4\right] \mathrm{d}r \\
&= \frac{\pi}{6}\left[-\frac{1}{5}(2-r^2)^{\frac{5}{2}} - \frac{r^5}{5}\right]_0^1 \\
&= \frac{\pi}{15}(2\sqrt{2}-1)
\end{aligned}$$

3. 计算下列三重积分.

(2) $\iiint_V \sqrt{x^2+y^2}\,\mathrm{d}x\mathrm{d}y\mathrm{d}z$, V: 由 $x^2+y^2=z^2$, $z=1$ 围成;

解. 该区域满足 $z \ge 0$.

采用柱面坐标变换: 记 $\begin{cases} x = r\cos\theta \\ y = r\sin\theta \\ z = z \end{cases}$ ，则积分区域为 $\begin{cases} 0 \le z \le 1 \\ 0 \le \theta \le 2\pi \\ 0 \le r \le z \end{cases}$ ，雅可比行列式为 $\dfrac{\partial(x,y,z)}{\partial(r,\theta,z)} = r$.

因此

$$\begin{aligned}
\iiint_V \sqrt{x^2+y^2}\,\mathrm{d}x\mathrm{d}y\mathrm{d}z &= \int_0^1 \mathrm{d}z \int_0^{2\pi} \mathrm{d}\theta \int_0^z r^2 \,\mathrm{d}r \\
&= \frac{1}{3}\int_0^1 \mathrm{d}z \int_0^{2\pi} z^3 \,\mathrm{d}\theta \\
&= \frac{2\pi}{3}\int_0^1 z^3 \,\mathrm{d}z \\
&= \frac{\pi}{6}
\end{aligned}$$

(4) $\iiint_V xyz\,\mathrm{d}x\mathrm{d}y\mathrm{d}z$, V: 是 $x^2+y^2+z^2\le 1$ 的第一卦限部分;

解. 采用球面坐标变换: 记 $\begin{cases} x=r\sin\theta\cos\varphi \\ y=r\sin\theta\sin\varphi \\ z=r\cos\theta \end{cases}$, 则积分区域为 $\begin{cases} 0\le r\le 1 \\ 0\le\theta\le\frac{\pi}{2} \\ 0\le\varphi\le\frac{\pi}{2} \end{cases}$, 雅可比行列式为 $\dfrac{\partial(x,y,z)}{\partial(r,\theta,\varphi)}=r^2\sin\theta$.

因此

$$\begin{aligned}\iiint_V xyz\,\mathrm{d}x\mathrm{d}y\mathrm{d}z &= \int_0^1 r^5\,\mathrm{d}r\int_0^{\frac{\pi}{2}}\sin^3\theta\cos\theta\,\mathrm{d}\theta\int_0^{\frac{\pi}{2}}\sin\varphi\cos\varphi\,\mathrm{d}\varphi \\ &= \frac{1}{48}\end{aligned}$$

(6) $\iiint_V |x^2+y^2+z^2-1|\,\mathrm{d}x\mathrm{d}y\mathrm{d}z$, V: $x^2+y^2+z^2\le 4$;

解. 由于被积函数关于原点中心对称, 可先计算第一卦限的积分再乘以 8.

采用球面坐标变换: 记 $\begin{cases} x=r\sin\theta\cos\varphi \\ y=r\sin\theta\sin\varphi \\ z=r\cos\theta \end{cases}$, 则积分区域为 $\begin{cases} 0\le r\le 2 \\ 0\le\theta\le\frac{\pi}{2} \\ 0\le\varphi\le\frac{\pi}{2} \end{cases}$, 雅可比行列式为 $\dfrac{\partial(x,y,z)}{\partial(r,\theta,\varphi)}=r^2\sin\theta$.

因此

$$\begin{aligned}\iiint_V |x^2+y^2+z^2-1|\,\mathrm{d}x\mathrm{d}y\mathrm{d}z &= 8\int_0^2\mathrm{d}r\int_0^{\frac{\pi}{2}}\mathrm{d}\theta\int_0^{\frac{\pi}{2}}\left|r^2-1\right|r^2\sin\theta\,\mathrm{d}\varphi \\ &= 8\left(\int_0^1(r^2-r^4)\,\mathrm{d}r+\int_1^2(r^4-r^2)\,\mathrm{d}r\right)\int_0^{\frac{\pi}{2}}\sin\theta\,\mathrm{d}\theta\int_0^{\frac{\pi}{2}}\mathrm{d}\varphi \\ &= 16\pi\end{aligned}$$

(8) $\iiint_V (|x|+z)\mathrm{e}^{-(x^2+y^2+z^2)}\,\mathrm{d}x\mathrm{d}y\mathrm{d}z$, V: $1\le x^2+y^2+z^2\le 4$.

解. 由于区域关于平面 $x=0$ 对称, 且被积函数中 $|x|$ 为偶函数, 可先计算 $x\ge 0$, $y\ge 0$ 部分的积分再乘以 4.

采用球面坐标变换: 记 $\begin{cases} x=r\sin\theta\cos\varphi \\ y=r\sin\theta\sin\varphi \\ z=r\cos\theta \end{cases}$, 则积分区域为 $\begin{cases} 1\le r\le 2 \\ 0\le\theta\le\pi \\ 0\le\varphi\le\frac{\pi}{2} \end{cases}$, 雅可比行列式为 $\dfrac{\partial(x,y,z)}{\partial(r,\theta,\varphi)}=r^2\sin\theta$.

因此

$$\begin{aligned}\iiint_V (|x|+z)\mathrm{e}^{-(x^2+y^2+z^2)}\,\mathrm{d}x\mathrm{d}y\mathrm{d}z &= 4\int_1^2 r^3\mathrm{e}^{-r^2}\,\mathrm{d}r\int_0^\pi\,\mathrm{d}\theta\int_0^{\frac{\pi}{2}}(\sin\theta\cos\varphi+\cos\theta)\sin\theta\,\mathrm{d}\varphi\\ &= 4\int_1^2 r^3\mathrm{e}^{-r^2}\,\mathrm{d}r\int_0^\pi\left(\sin^2\theta\cdot\frac{\pi}{2}+\frac{\pi}{2}\sin\theta\cos\theta\right)\,\mathrm{d}\theta\\ &= 2\pi\int_1^2 r^3\mathrm{e}^{-r^2}\,\mathrm{d}r\int_0^\pi\sin^2\theta\,\mathrm{d}\theta\\ &= 2\pi\cdot\frac{2\mathrm{e}^3-5}{2\mathrm{e}^4}\cdot\frac{\pi}{2}\\ &= \frac{\pi^2(2\mathrm{e}^3-5)}{2\mathrm{e}^4}\end{aligned}$$

4. 利用对称性求下列三重积分.

(2) $\iiint_V x\,\mathrm{d}x\mathrm{d}y\mathrm{d}z$, V: 由 $x^2+y^2=z^2$, $x^2+y^2=1$ 围成;

解. 采用柱面坐标变换: 记 $\begin{cases}x=r\cos\theta\\ y=r\sin\theta\\ z=z\end{cases}$, 则积分区域为 $\begin{cases}0\le r\le 1\\ 0\le\theta\le 2\pi\\ -r\le z\le r\end{cases}$, 雅可比行列式为 $\dfrac{\partial(x,y,z)}{\partial(r,\theta,z)}=r$.

由于该区域为对称双锥, 且被积函数 x 关于 x 为奇函数, 因此

$$\begin{aligned}\iiint_V x\,\mathrm{d}x\mathrm{d}y\mathrm{d}z &= \int_0^1 r^2\,\mathrm{d}r\int_0^{2\pi}\cos\theta\,\mathrm{d}\theta\int_{-r}^{r}\,\mathrm{d}z\\ &= 0\end{aligned}$$

(4) $\iiint_V (x^2+y^2)\,\mathrm{d}x\mathrm{d}y\mathrm{d}z$, V: $r^2\le x^2+y^2+z^2\le R^2$, $z\ge 0$;

解. 采用球面坐标变换: 记 $\begin{cases}x=\rho\sin\theta\cos\varphi\\ y=\rho\sin\theta\sin\varphi\\ z=\rho\cos\theta\end{cases}$, 则积分区域为 $\begin{cases}r\le\rho\le R\\ 0\le\theta\le\frac{\pi}{2}\\ 0\le\varphi\le 2\pi\end{cases}$, 雅可比行列式为 $\dfrac{\partial(x,y,z)}{\partial(\rho,\theta,\varphi)}=\rho^2\sin\theta$.

因此

$$\begin{aligned}\iiint_V (x^2+y^2)\,\mathrm{d}x\mathrm{d}y\mathrm{d}z &= \int_r^R \rho^4\,\mathrm{d}\rho\int_0^{\frac{\pi}{2}}\sin^3\theta\,\mathrm{d}\theta\int_0^{2\pi}\,\mathrm{d}\varphi\\ &= \frac{2\pi}{5}\cdot(R^5-r^5)\cdot\left(-\cos\theta+\frac{1}{3}\cos^3\theta\right)\Big|_0^{\frac{\pi}{2}}\\ &= \frac{4\pi}{15}(R^5-r^5)\end{aligned}$$

(6) $\iiint_V \frac{z\ln(x^2+y^2+z^2+1)}{x^2+y^2+z^2+1}\,\mathrm{d}x\mathrm{d}y\mathrm{d}z$, V: $x^2+y^2+z^2\le 1$;

解. 注意到被积函数关于 z 为奇函数, 而积分区域关于平面 $z=0$ 对称, 因此原积分值为 0.

验证: 采用球面坐标变换, 记 $\begin{cases} x=r\sin\theta\cos\varphi \\ y=r\sin\theta\sin\varphi \\ z=r\cos\theta \end{cases}$, 则积分区域为 $\begin{cases} 0\le r\le 1 \\ 0\le\theta\le\pi \\ 0\le\varphi\le 2\pi \end{cases}$, 雅可比行列式为 $\frac{\partial(x,y,z)}{\partial(r,\theta,\varphi)}=r^2\sin\theta$.

因此

$$\iiint_V \frac{z\ln(x^2+y^2+z^2+1)}{x^2+y^2+z^2+1}\,\mathrm{d}x\mathrm{d}y\mathrm{d}z=\int_0^1\frac{r^3\ln(r^2+1)}{r^2+1}\,\mathrm{d}r\int_0^\pi\cos\theta\sin\theta\,\mathrm{d}\theta\int_0^{2\pi}\,\mathrm{d}\varphi,$$

其中 $\int_0^\pi\cos\theta\sin\theta\,\mathrm{d}\theta=0$.

因此 $\iiint_V \frac{z\ln(x^2+y^2+z^2+1)}{x^2+y^2+z^2+1}\,\mathrm{d}x\mathrm{d}y\mathrm{d}z=0$.

(8) 书上没这道题目.

5. 计算下列曲面围成的立体体积.

(1) $y=0$, $z=0$, $3x+y=6$, $3x+2y=12$, $x+y+z=6$;

解. 积分区域可表示为 $\begin{cases} 0\le y\le 6 \\ 2-\frac{y}{3}\le x\le 4-\frac{2y}{3} \\ 0\le z\le 6-x-y \end{cases}$.

因此体积为

$$\begin{aligned}
\int_0^6\,\mathrm{d}y\int_{2-\frac{y}{3}}^{4-\frac{2y}{3}}\,\mathrm{d}x\int_0^{6-x-y}\,\mathrm{d}z &= \int_0^6\,\mathrm{d}y\int_{2-\frac{y}{3}}^{4-\frac{2y}{3}}(6-x-y)\,\mathrm{d}x \\
&= \int_0^6\left(6x-\frac{x^2}{2}-yx\right)\Big|_{2-\frac{y}{3}}^{4-\frac{2y}{3}}\,\mathrm{d}y \\
&= \int_0^6\left(\frac{y^2}{6}-2y+6\right)\,\mathrm{d}y \\
&= \left(\frac{y^3}{18}-y^2+6y\right)\Big|_0^6 \\
&= 12
\end{aligned}$$

(3) $z^2+x^2=1$ 和 $x+y+z=3$, $y=0$;

解. 采用柱面坐标变换: 记 $\begin{cases} x=r\cos\theta \\ z=r\sin\theta \\ y=y \end{cases}$, 则积分区域为 $\begin{cases} 0\le r\le 1 \\ 0\le\theta\le 2\pi \\ 0\le y\le 3-r(\cos\theta+\sin\theta) \end{cases}$, 雅可比行列式为 $\frac{\partial(x,y,z)}{\partial(r,\theta,y)}=r$.

因此体积为

$$\begin{aligned}\int_0^1 \mathrm{d}r \int_0^{2\pi} (3 - r(\cos\theta + \sin\theta))r \,\mathrm{d}\theta &= \int_0^1 \left(3\theta - r\sin\theta + r\cos\theta\right) r\big|_0^{2\pi} \,\mathrm{d}r \\ &= \int_0^1 6\pi r \,\mathrm{d}r \\ &= 3\pi\end{aligned}$$

(5) $\frac{x^2}{9} + \frac{y^2}{4} = 1$, $z = xy$ (在第一卦限部分);

解. 采用广义柱面坐标变换: 记 $\begin{cases} x = 3r\cos\theta \\ y = 2r\sin\theta \\ z = z \end{cases}$, 则积分区域为 $\begin{cases} 0 \le r \le 1 \\ 0 \le \theta \le \frac{\pi}{2} \\ 0 \le z \le 6r^2\cos\theta\sin\theta \end{cases}$, 雅可比行列式为 $\dfrac{\partial(x,y,z)}{\partial(r,\theta,z)} = 6r$.

因此体积为

$$36\int_0^1 r^3 \,\mathrm{d}r \int_0^{\frac{\pi}{2}} \cos\theta\sin\theta \,\mathrm{d}\theta = \frac{9}{2}$$

(7) $\frac{x^2}{a^2} + \frac{y^2}{b^2} + \frac{z^2}{c^2} = 1$, $\frac{x^2}{a^2} + \frac{y^2}{b^2} = \frac{z^2}{c^2}$ (含 z 轴部分).

解. 采用广义球面坐标变换: 记 $\begin{cases} x = ar\sin\theta\cos\varphi \\ y = br\sin\theta\sin\varphi \\ z = cr\cos\theta \end{cases}$, 则积分区域为 $\begin{cases} 0 \le r \le 1 \\ 0 \le \theta \le \pi \\ 0 \le \varphi \le 2\pi \end{cases}$, 雅可比行列式为 $\dfrac{\partial(x,y,z)}{\partial(r,\theta,\varphi)} = abcr^2\sin\theta$.

由于该区域包含 z 轴, 需满足 $\sin^2\theta \le \cos^2\theta$, 解得 $0 \le \theta \le \dfrac{\pi}{4}$ 或 $\dfrac{3\pi}{4} \le \theta \le \pi$.

由于两个区域体积相等, 可先计算一个区域的体积再乘以 2.

因此体积为

$$\begin{aligned}2abc \cdot 2\pi \int_0^1 r^2 \,\mathrm{d}r \int_0^{\frac{\pi}{4}} \sin\theta \,\mathrm{d}\theta &= \frac{4}{3}abc\pi \left(1 - \frac{\sqrt{2}}{2}\right) \\ &= \frac{4 - 2\sqrt{2}}{3}abc\pi\end{aligned}$$

7. 设 $F(t) = \displaystyle\iiint\limits_{x^2+y^2+z^2 \le t^2} f(x^2+y^2+z^2) \,\mathrm{d}x\mathrm{d}y\mathrm{d}z$, 其中 f 为可微分函数, 求 $F'(t)$.

解. 采用球面坐标变换: 记 $\begin{cases} x = r\sin\theta\cos\varphi \\ y = r\sin\theta\sin\varphi \\ z = r\cos\theta \end{cases}$, 则积分区域为 $\begin{cases} 0 \le r \le t \\ 0 \le \theta \le \pi \\ 0 \le \varphi \le 2\pi \end{cases}$, 雅可比行列式为 $\dfrac{\partial(x,y,z)}{\partial(r,\theta,\varphi)} = r^2\sin\theta$.

因此

$$\iiint\limits_{x^2+y^2+z^2\le t^2} f(x^2+y^2+z^2)\,\mathrm{d}x\mathrm{d}y\mathrm{d}z = \int_0^t f(r^2)r^2\,\mathrm{d}r\int_0^\pi \sin\theta\,\mathrm{d}\theta\int_0^{2\pi}\,\mathrm{d}\varphi$$
$$= 4\pi\int_0^t f(r^2)r^2\,\mathrm{d}r$$

由变上限积分求导法则可得 $F'(t) = 4\pi t^2 f(t^2)$.

8. 证明: $\displaystyle\iiint\limits_{x^2+y^2+z^2\le 1} f(z)\,\mathrm{d}v = \pi\int_{-1}^1 f(z)(1-z^2)\,\mathrm{d}z$.

证. 考虑等式左边的三重积分, 先对 x, y 积分.

对于固定的 z, 截面区域为 $x^2+y^2 \le 1-z^2$. 采用柱面坐标变换: 记 $\begin{cases} x = r\cos\theta \\ y = r\sin\theta \end{cases}$, 则 $\begin{cases} 0 \le r \le \sqrt{1-z^2} \\ 0 \le \theta \le 2\pi \end{cases}$, 雅可比行列式为 $\dfrac{\partial(x,y)}{\partial(r,\theta)} = r$.

因此

$$\iiint\limits_{x^2+y^2+z^2\le 1} f(z)\,\mathrm{d}x\mathrm{d}y\mathrm{d}z = \int_{-1}^1 f(z)\,\mathrm{d}z\iint\limits_{x^2+y^2\le 1-z^2}\mathrm{d}x\mathrm{d}y$$
$$= \int_{-1}^1 f(z)\,\mathrm{d}z\int_0^{\sqrt{1-z^2}} r\,\mathrm{d}r\int_0^{2\pi}\,\mathrm{d}\theta$$
$$= \pi\int_{-1}^1 f(z)(1-z^2)\,\mathrm{d}z$$

这与等式右边相等.

Q.E.D.

12. 一个物体是由两个半径各为 R 和 r $(R > r)$ 的同心球所围成, 已知材料的密度和到球心的距离成反比, 且在距离等于 1 处的密度等于 k, 求物体的总质量.

解. 采用球面坐标变换: 记 $\begin{cases} x = \rho\sin\theta\cos\varphi \\ y = \rho\sin\theta\sin\varphi \\ z = \rho\cos\theta \end{cases}$, 则体积元为 $\rho^2\sin\theta\,\mathrm{d}\rho\mathrm{d}\theta\mathrm{d}\varphi$.

由题意可知, 在点 (ρ,θ,φ) 处的密度为 $\dfrac{k}{\rho}$.

因此物体的总质量为

$$2\pi\int_r^R \frac{k}{\rho}\cdot\rho^2\,\mathrm{d}\rho\int_0^\pi \sin\theta\,\mathrm{d}\theta = 2k\pi(R^2-r^2)$$

13. 半径为 a 的圆盘, 其各点的密度和到圆心的距离成正比 (比例系数为 1), 今内切于圆盘截去半径为 $\frac{a}{2}$ 之小圆, 求余下部分的重心坐标.

解. 由对称性可知, 重心位于两圆圆心连成的直线上. 以该直线为 x 轴, 大圆圆心为坐标原点, 垂直方向为 y 轴建立坐标系, 则只需计算重心在 x 轴方向上的坐标.

坐标 (x,y) 处的密度为 $\sqrt{x^2+y^2}$.

需要计算剩余部分的力矩

$$\iint\limits_{x^2+y^2\le a^2} x\sqrt{x^2+y^2}\,\mathrm{d}x\mathrm{d}y - \iint\limits_{\left(x-\frac{a}{2}\right)^2+y^2\le\frac{a^2}{4}} x\sqrt{x^2+y^2}\,\mathrm{d}x\mathrm{d}y$$

和质量

$$\iint\limits_{x^2+y^2\le a^2} \sqrt{x^2+y^2}\,\mathrm{d}x\mathrm{d}y - \iint\limits_{\left(x-\frac{a}{2}\right)^2+y^2\le\frac{a^2}{4}} \sqrt{x^2+y^2}\,\mathrm{d}x\mathrm{d}y$$

采用柱面坐标变换: 记 $\begin{cases} x = r\cos\theta \\ y = r\sin\theta \end{cases}$, 则雅可比行列式为 $\dfrac{\partial(x,y)}{\partial(r,\theta)} = r$.

对于大圆区域, 积分区域为 $\begin{cases} 0\le r\le a \\ 0\le\theta\le 2\pi \end{cases}$.

对于小圆区域, 由 $\left(x-\dfrac{a}{2}\right)^2+y^2\le\dfrac{a^2}{4}$ 可得 $r^2-ar\cos\theta+\dfrac{a^2}{4}\le\dfrac{a^2}{4}$, 即 $r\le a\cos\theta$. 因此积分区域为 $\begin{cases} -\frac{\pi}{2}\le\theta\le\frac{\pi}{2} \\ 0\le r\le a\cos\theta \end{cases}$.

首先计算大圆的力矩和质量:

$$\begin{aligned}
\text{力矩} &= \iint\limits_{x^2+y^2\le a^2} x\sqrt{x^2+y^2}\,\mathrm{d}x\mathrm{d}y \\
&= \int_0^a r^3\,\mathrm{d}r\int_0^{2\pi}\cos\theta\,\mathrm{d}\theta \\
&= \frac{a^4}{4}\cdot 0 \\
&= 0
\end{aligned}$$

这是符合直觉的.

$$
\begin{aligned}
\text{质量} &= \iint\limits_{x^2+y^2\le a^2} \sqrt{x^2+y^2}\,\mathrm{d}x\mathrm{d}y \\
&= \int_0^a r^2\,\mathrm{d}r \int_0^{2\pi}\,\mathrm{d}\theta \\
&= \frac{2a^3\pi}{3}
\end{aligned}
$$

接下来计算小圆的力矩:

$$
\begin{aligned}
\text{力矩} &= \iint\limits_{\left(x-\frac{a}{2}\right)^2+y^2\le\frac{a^2}{4}} x\sqrt{x^2+y^2}\,\mathrm{d}x\mathrm{d}y \\
&= \int_{-\frac{\pi}{2}}^{\frac{\pi}{2}}\,\mathrm{d}\theta \int_0^{a\cos\theta} r^3\cos\theta\,\mathrm{d}r \\
&= \frac{a^4}{4}\int_{-\frac{\pi}{2}}^{\frac{\pi}{2}}\cos^5\theta\,\mathrm{d}\theta
\end{aligned}
$$

计算 $\displaystyle\int_{-\frac{\pi}{2}}^{\frac{\pi}{2}}\cos^5\theta\,\mathrm{d}\theta$: 由于被积函数为偶函数, 因此

$$
\begin{aligned}
\int_{-\frac{\pi}{2}}^{\frac{\pi}{2}}\cos^5\theta\,\mathrm{d}\theta &= 2\int_0^{\frac{\pi}{2}}\cos^5\theta\,\mathrm{d}\theta \\
&= 2\int_0^1 (1-u^2)^2\,\mathrm{d}u \\
&= 2\int_0^1 (1-2u^2+u^4)\,\mathrm{d}u \\
&= 2\left(u-\frac{2}{3}u^3+\frac{1}{5}u^5\right)\Bigg|_0^1 \\
&= \frac{16}{15}
\end{aligned}
$$

其中采用了换元法: 记 $u=\sin\theta$, 则 $\mathrm{d}u=\cos\theta\,\mathrm{d}\theta$.
因此小圆的力矩为 $\dfrac{4a^4}{15}$.
再计算小圆的质量:

$$
\begin{aligned}
\text{质量} &= \iint\limits_{\left(x-\frac{a}{2}\right)^2+y^2\le\frac{a^2}{4}} \sqrt{x^2+y^2}\,\mathrm{d}x\mathrm{d}y \\
&= \int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} \mathrm{d}\theta \int_0^{a\cos\theta} r^2\,\mathrm{d}r \\
&= \frac{a^3}{3}\int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} \cos^3\theta\,\mathrm{d}\theta \\
&= \frac{4a^3}{9}
\end{aligned}
$$

因此剩余部分的重心 x 坐标为

$$
\frac{0-\frac{4a^4}{15}}{\frac{2a^3\pi}{3}-\frac{4a^3}{9}} = \frac{6a}{10-15\pi}
$$

即重心坐标为 $\left(\dfrac{6a}{10-15\pi}, 0\right)$.

第 10 章综合习题

2. 计算三重积分

$$
I = \iiint_{[0,1]^3} \frac{\mathrm{d}u\mathrm{d}v\mathrm{d}w}{(1+u^2+v^2+w^2)^2}.
$$

解. 令 $\begin{cases} u = r\sin\theta \\ v = r\cos\theta \\ w = w \end{cases}$，所以 $0\le\theta\le\frac{\pi}{2}$, $\begin{cases} 0\le r\le\frac{1}{\cos\theta} & 0\le\theta\le\frac{\pi}{4} \\ 0\le r\le\frac{1}{\sin\theta} & \frac{\pi}{4}\le\theta\le\frac{\pi}{2} \end{cases}$, 这两部分是对称的，只计算其中一部分并 $\times 2$ 即可。$\frac{\partial(u,v,w)}{\partial(r,\theta,w)} = r$

所以

$$
\begin{aligned}
\iiint_{[0,1]^3} \frac{\mathrm{d}u\mathrm{d}v\mathrm{d}w}{(1+u^2+v^2+w^2)^2} &= 2\int_0^{\frac{\pi}{4}} \mathrm{d}\theta \int_0^1 \mathrm{d}w \int_0^{\frac{1}{\cos\theta}} \frac{r}{(1+r^2+w^2)^2}\,\mathrm{d}r \\
&= \int_0^{\frac{\pi}{4}} \mathrm{d}\theta \int_0^1 \left(\frac{1}{1+w^2} - \frac{1}{1+\frac{1}{\cos^2\theta}+w^2}\right) \mathrm{d}w
\end{aligned}
$$

其中 $\int_0^1 \frac{1}{1+w^2}\,\mathrm{d}w = \arctan w|_0^1 = \frac{\pi}{4}$

计算 $\int_0^1 \frac{1}{1+\frac{1}{\cos^2\theta}+w^2}\,\mathrm{d}w$:

令 $a = \sqrt{1+\frac{1}{\cos^2\theta}}$, $w = at$, 所以 $\mathrm{d}w = a\,\mathrm{d}t$

所以

$$\begin{aligned}\int_0^1 \frac{1}{1+\frac{1}{\cos^2\theta}+w^2}\,\mathrm{d}w &= \int_0^{\frac{1}{a}} \frac{a}{a^2+a^2t^2}\,\mathrm{d}t\\ &= \frac{1}{a}\int_0^{\frac{1}{a}} \frac{1}{1+t^2}\,\mathrm{d}t\\ &= \frac{1}{a}\arctan\frac{1}{a}\\ &= \frac{1}{\sqrt{1+\frac{1}{\cos^2\theta}}}\arctan\frac{1}{\sqrt{1+\frac{1}{\cos^2\theta}}}\\ &= \frac{\cos\theta}{\sqrt{1+\cos^2\theta}}\arctan\frac{\cos\theta}{\sqrt{1+\cos^2\theta}}\end{aligned}$$

所以 $\int_0^1\left(\frac{1}{1+w^2}-\frac{1}{1+\frac{1}{\cos^2\theta}+w^2}\right)\mathrm{d}w = \frac{\pi}{4}-\frac{\cos\theta}{\sqrt{1+\cos^2\theta}}\arctan\frac{\cos\theta}{\sqrt{1+\cos^2\theta}}$

所以

$$\begin{aligned}\iiint_{[0,1]^3}\frac{\mathrm{d}u\mathrm{d}v\mathrm{d}w}{(1+u^2+v^2+w^2)^2} &= \int_0^{\frac{\pi}{4}}(\frac{\pi}{4}-\frac{\cos\theta}{\sqrt{1+\cos^2\theta}}\arctan\frac{\cos\theta}{\sqrt{1+\cos^2\theta}})\,\mathrm{d}\theta\\ &= \frac{\pi^2}{16}-\int_0^{\frac{\pi}{4}}\frac{\cos\theta}{\sqrt{1+\cos^2\theta}}\arctan\frac{\cos\theta}{\sqrt{1+\cos^2\theta}}\,\mathrm{d}\theta\\ &= \frac{\pi^2}{32}\end{aligned}$$

3. 设 $a>0, b>0$. 试求下面的积分:

(1) $I_1=\int_0^1 \sin\left(\ln\frac{1}{x}\right)\cdot\frac{x^b-x^a}{\ln x}\,\mathrm{d}x$;

解. 注意到 $\frac{x^b-x^a}{\ln x}=\int_a^b x^y\,\mathrm{d}y$

所以 $I_1=\int_0^1\sin\left(\ln\frac{1}{x}\right)\cdot\frac{x^b-x^a}{\ln x}\,\mathrm{d}x=-\int_a^b\mathrm{d}y\int_0^1\sin(\ln x)\cdot x^y\,\mathrm{d}x$

令 $c=\ln x$, 所以 $\mathrm{d}c=\frac{1}{x}\mathrm{d}x$

所以 $\int_a^b\mathrm{d}y\int_0^1\sin(\ln x)\cdot x^y\,\mathrm{d}x=\int_a^b\mathrm{d}y\int_{-\infty}^0\sin c\cdot\mathrm{e}^{c(y+1)}\,\mathrm{d}c$

其中

$$\begin{aligned}\int\sin c\cdot\mathrm{e}^{c(y+1)}\,\mathrm{d}c &= -\cos c\cdot\mathrm{e}^{c(y+1)}+(y+1)\int\cos c\cdot\mathrm{e}^{c(y+1)}\,\mathrm{d}c\\ &= -(y+1)^2\int\sin c\cdot\mathrm{e}^{c(y+1)}\,\mathrm{d}c-(\cos c-(y+1)\sin c)\mathrm{e}^{c(y+1)}\end{aligned}$$

所以 $\int\sin c\cdot\mathrm{e}^{c(y+1)}\,\mathrm{d}c=-\frac{(\cos c-(y+1)\sin c)\mathrm{e}^{c(y+1)}}{(y+1)^2+1}+C$

所以 $\int_{-\infty}^0\sin c\cdot\mathrm{e}^{c(y+1)}\,\mathrm{d}c=-\frac{1}{1+(y+1)^2}$

所以

$$
\begin{aligned}
\int_0^1 \sin\left(\ln\frac{1}{x}\right)\cdot\frac{x^b-x^a}{\ln x}\,\mathrm{d}x &= -\int_a^b \mathrm{d}y\int_0^1 \sin(\ln x)\cdot x^y\,\mathrm{d}x \\
&= \int_a^b \frac{1}{1+(y+1)^2}\,\mathrm{d}y \\
&= \arctan(y+1)\big|_a^b \\
&= \arctan(b+1)-\arctan(a+1)
\end{aligned}
$$

(2) $I_2=\displaystyle\int_0^1 \cos\left(\ln\frac{1}{x}\right)\cdot\frac{x^b-x^a}{\ln x}\,\mathrm{d}x.$

解. 令 $c=\ln x$, 所以原式 $\int_0^1 \cos\left(\ln\frac{1}{x}\right)\cdot\frac{x^b-x^a}{\ln x}\,\mathrm{d}x=\int_a^b \mathrm{d}y\int_{-\infty}^0 \cos c\cdot \mathrm{e}^{c(y+1)}\,\mathrm{d}c$

其中

$$
\begin{aligned}
\int \cos c\cdot\mathrm{e}^{c(y+1)}\,\mathrm{d}c &= \sin c\cdot\mathrm{e}^{c(y+1)}-(y+1)\int \sin c\cdot\mathrm{e}^{c(y+1)}\,\mathrm{d}c \\
&= (\sin c-(y+1)\cos c)\mathrm{e}^{c(y+1)}-(y+1)^2\int\cos c\cdot\mathrm{e}^{c(y+1)}\,\mathrm{d}c
\end{aligned}
$$

所以 $\int \cos c\cdot\mathrm{e}^{c(y+1)}\,\mathrm{d}c=\frac{(\sin c-(y+1)\cos c)\mathrm{e}^{c(y+1)}}{(y+1)^2+1}+C$

所以 $\int_{-\infty}^0 \cos c\cdot\mathrm{e}^{c(y+1)}\,\mathrm{d}c=\frac{y+1}{(y+1)^2+1}$

所以 $\int\frac{y+1}{1+(y+1)^2}\,\mathrm{d}y=\frac{1}{2}\ln(1+(y+1)^2)+C$

所以

$$
\begin{aligned}
\int_0^1 \cos\left(\ln\frac{1}{x}\right)\cdot\frac{x^b-x^a}{\ln x}\,\mathrm{d}x &= \int_a^b\frac{y+1}{1+(y+1)^2}\,\mathrm{d}y \\
&= \frac{1}{2}\ln\frac{1+(b+1)^2}{1+(a+1)^2}
\end{aligned}
$$

4. 设 $D=\{(x,y)\mid x^2+y^2\le 1\}$. 求 $I=\displaystyle\iint\limits_D\left|\frac{x+y}{\sqrt{2}}-x^2-y^2\right|\mathrm{d}x\mathrm{d}y.$

解. 令 $\begin{cases}u=\frac{x+y}{\sqrt{2}}\\ v=\frac{x-y}{\sqrt{2}}\end{cases}$, 所以 $u^2+v^2\le 1$, $\frac{\partial(x,y)}{\partial(u,v)}=1$

所以即求 $\iint_{u^2+v^2\le 1}|u-u^2-v^2|\,\mathrm{d}u\mathrm{d}v$.

令 $\begin{cases}u=r\cos\theta\\ v=r\sin\theta\end{cases}$, 所以 $\begin{cases}0\le r\le 1\\ 0\le\theta\le 2\pi\end{cases}$, $\frac{\partial(u,v)}{\partial(r,\theta)}=r$

所以 $\iint_{u^2+v^2\le 1}|u-u^2-v^2|\,\mathrm{d}u\mathrm{d}v=\int_0^{2\pi}\mathrm{d}\theta\int_0^1 r^2\,|\cos\theta-r|\,\mathrm{d}r$

令 $c=\cos\theta$, 则 $\begin{cases}c\ge 0 & -\frac{\pi}{2}\le\theta\le\frac{\pi}{2}\\ c\le 0 & \frac{\pi}{2}\le\theta\le\frac{3\pi}{2}\end{cases}$

当 $c \geq 0$,

$$
\begin{aligned}
\int_0^1 r^2|\cos\theta - r|\,\mathrm{d}r &= \int_0^c r^2(c-r)\,\mathrm{d}r + \int_c^1 r^2(r-c)\,\mathrm{d}r \\
&= \int_0^c (cr^2 - r^3)\,\mathrm{d}r + \int_c^1 (r^3 - cr^2)\,\mathrm{d}r \\
&= \left(\frac{cr^3}{3} - \frac{r^4}{4}\right)\Bigg|_0^c + \left(\frac{r^4}{4} - \frac{cr^3}{3}\right)\Bigg|_c^1 \\
&= \frac{c^4}{3} - \frac{c^4}{4} + \left(\frac{1}{4} - \frac{c}{3}\right) - \left(\frac{c^4}{4} - \frac{c^4}{3}\right) \\
&= \frac{c^4}{12} + \frac{1}{4} - \frac{c}{3} + \frac{c^4}{12} \\
&= \frac{1}{4} - \frac{c}{3} + \frac{c^4}{6}
\end{aligned}
$$

当 $c \leq 0$, $\int_0^1 r^2|\cos\theta - r|\,\mathrm{d}r = \int_0^1 (r^3 - cr^2)\,\mathrm{d}r = \left(\frac{r^4}{4} - \frac{cr^3}{3}\right)\Big|_0^1 = \frac{1}{4} - \frac{c}{3}$
所以

$$
\begin{aligned}
\iint\limits_D \left|\frac{x+y}{\sqrt{2}} - x^2 - y^2\right| \mathrm{d}x\mathrm{d}y &= \int_0^{2\pi} \mathrm{d}\theta \int_0^1 r^2\left|\cos\theta - r\right|\,\mathrm{d}r \\
&= 2\int_0^{\pi/2} \left(\frac{1}{4} - \frac{\cos\theta}{3} + \frac{\cos^4\theta}{6}\right)\mathrm{d}\theta + \int_{\pi/2}^{3\pi/2} \left(\frac{1}{4} - \frac{\cos\theta}{3}\right)\mathrm{d}\theta \\
&= \frac{\pi}{4} - \frac{2}{3} + \frac{\pi}{16} + \frac{\pi}{4} + \frac{2}{3} \\
&= \frac{9\pi}{16}
\end{aligned}
$$